



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PART TEST

13-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	B	A	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	C	A,B,C,D	A,D	B,C	A,B,C,D	A,B
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	53.33	120.00	125.60	1.55 to 1.64	75.00	30.00
SECTION-II (ii)	Q.	7	8	9			
	A.	19	6	8			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	A	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,D	B,D	A,B,C	A,B,C	B,D	A,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	11.00	14.08 to 14.10	88.80 to 89.00	0.60	2.25	1.11
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	150			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	A	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,D	A,B	C,D	A,D	A,C	B
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.00	0.00	4.00	4.00	7.00	0.00
SECTION-II (ii)	Q.	7	8	9			
	A.	1	0	2			

HINT – SHEET

PART-1 : PHYSICS

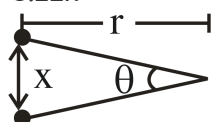
SECTION-I (i)

1. Ans (A)

$$\theta \geq \frac{1.22\lambda}{D}$$

$$\frac{x}{r} \geq \frac{1.22\lambda}{D}$$

$$\frac{x \times D}{1.22\lambda} \geq r$$



$$r_{\max} = \frac{x D}{1.22\lambda} \text{ maximum distance from where}$$

dots can be seen separately

Wavelength for which $r_{\max}$ is least will appear merged first

So red colour dots will merge first.

2. Ans (B)

$\Delta x_1 \rightarrow$ Path difference before 'S₁' and 'S₂'

$\Delta x_2 \rightarrow$ Path difference after 'S₁' and 'S₂'

For central Max. to be at 'O', $\Delta x_1 = \Delta x_2$

$$\Delta x_1 = \sqrt{\frac{D^2}{4} + d^2} - \frac{D}{2}$$

$$= \frac{D}{2} \left(1 + \frac{4d^2}{D^2} \right)^{\frac{1}{2}} - \frac{D}{2}$$

$$= \frac{D}{2} \left(1 + \frac{2d^2}{D^2} \right) - \frac{D}{2}$$

$$= \frac{d^2}{D}$$

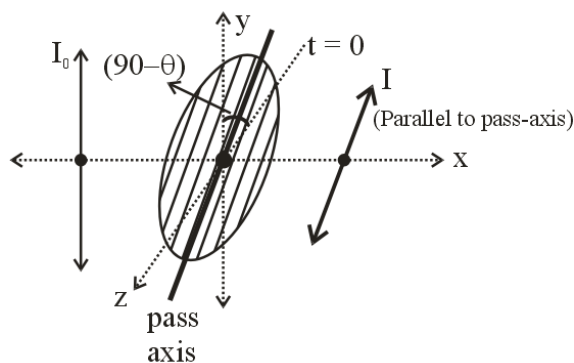
$$\Delta x_2 = (\mu - 1)t$$

$$\therefore (\mu - 1)t = \frac{d^2}{D}$$

$$\Rightarrow t = \frac{d^2}{(\mu - 1)D}$$

3. Ans (A)

At any time 't'



$$I = I_0 \cos^2(90 - \theta) \Rightarrow I = I_0 \sin^2 \theta$$

$$\therefore \frac{dI}{dt} = I_0 2 \sin \theta \cos \theta \frac{d\theta}{dt}$$

$$\Rightarrow \frac{dI}{dt} = I_0 \omega \sin 2\omega t \quad [\because \theta = \omega t]$$

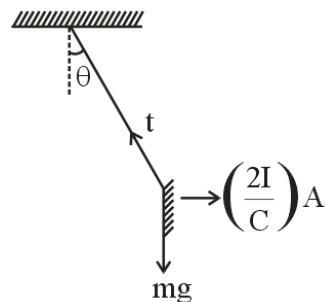
$$\left(\frac{dI}{dt} \right)_{\max} = I_0 \omega \text{ and}$$

$$\text{Angular frequency of } \frac{dI}{dt} = 2\omega$$

$$\therefore \text{Ratio} = \frac{I_0 \omega}{2\omega} = \frac{I_0}{2}$$

4. Ans (B)

$$T \cos \theta = mg \text{ and } T \sin \theta = \left(\frac{2I}{C} \right) A$$



$$\therefore T^2 \sin^2 \theta + T^2 \cos^2 \theta = (mg)^2 + \left(\frac{2IA}{C} \right)^2$$

$$\Rightarrow T^2 = (mg)^2 + \left(\frac{2 \times \frac{B_0^2}{2\mu_0} \times C \times A}{C} \right)^2$$

$$\Rightarrow T = \sqrt{(mg)^2 + \left(\frac{B_0^2 A}{\mu_0} \right)^2}$$

PART-1 : PHYSICS

SECTION-I (ii)

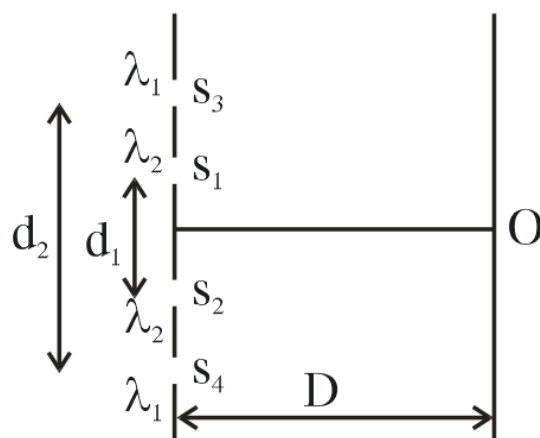
5. Ans (C)

Intensity of y distance

$$= 2I_0 \cos^2 \left(\frac{1}{2} \times \frac{2\pi}{\lambda_2} \times \frac{yd_1}{D} \right)$$

$$+ I_0 \cos^2 \left(\frac{1}{2} \times \frac{2\pi}{\lambda_1} \times \frac{yd_2}{D} \right)$$

$$I = 2I_0 \cos^2(5\pi y) + I_2 \cos^2(10\pi y)$$



$$\frac{dI}{dy} = 0$$

$$\sin(10\pi y) + \sin(20\pi y) = 0$$

$$\sin 10\pi y = 0 \Rightarrow 10\pi y = n\pi$$

and

$$\cos 10\pi y = -\frac{1}{2}$$

$$10\pi y = (2n+1)\pi \pm \frac{\pi}{3}$$

$$\text{maxima at } y = 0, \frac{1}{10}, \frac{2}{10}, \frac{3}{10}, \dots m$$

$$\text{minima at } y = \frac{2}{30}, \frac{4}{30}, \frac{8}{30}, \frac{10}{30}, \dots m$$

6. **Ans (A,B,C,D)**

Path difference

$$\Delta x = 2d\sqrt{2} - d\sqrt{2} = d\sqrt{2},$$

$$\text{where } d = \sqrt{2}\text{cm}$$

$$\Delta x = 2 \text{ cm.}$$

there is no net phase difference due to reflection

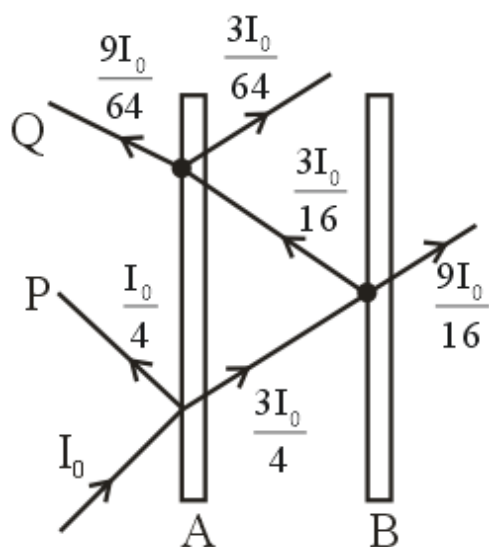
$$\text{So, } \phi = \frac{2\pi}{\lambda} \Delta x$$

$$\text{For maxima } \phi = 2n\pi \rightarrow \Delta x = n\lambda$$

$$\text{But } n\lambda = 2\text{cm. So, } n\lambda = \frac{2\text{cm}}{n}$$

$$\text{for } n = 1, \lambda = 2 \text{ cm and } f = 15 \text{ GHz}$$

$$\text{for } n = 2, \lambda = 1 \text{ cm and } f = 30 \text{ GHz}$$



Interference of radiation P and Q will have maxima of intensity.

$$I = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2} = \frac{49}{64}I_0$$

Minimum intensity

$$I = I_1 + I_2 - 2\sqrt{I_1}\sqrt{I_2} = \frac{I_0}{64}$$

7. **Ans (A,D)**

$$f_0 = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$xf_0 = \frac{1}{2 \times 0.7\ell} \sqrt{\frac{1.96T}{\mu}}$$

$$= \frac{1.4}{0.7} \times \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$xf_0 = 2f_0$$

$$x = 2$$

$$yf_0 = \frac{1}{2 \times 1.2\ell} \sqrt{\frac{0.64T}{\mu}}$$

$$= \frac{0.8}{1.2} \times \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$yf_0 = \frac{2}{3}f_0$$

$$y = \frac{2}{3}$$

$$xy = 2 \times \frac{2}{3} = \frac{4}{3}$$

$$\frac{x}{y} = \frac{2}{\frac{2}{3}} = 3$$

8. **Ans (B,C)**

$$\text{Velocity of wave} = \sqrt{\frac{T}{\mu}}$$

$$= \sqrt{\frac{T}{\gamma A}}$$

$$= \sqrt{\frac{yA\Delta\ell}{\ell} \times \frac{1}{\rho A}}$$

$$= \sqrt{\frac{y\Delta\ell}{\ell\rho}}$$

$$= \sqrt{\frac{45 \times 10^{10} \times 12.5 \times 10^{-3}}{5 \times 4.5 \times 10^3}}$$

$$= 500 \text{ m/s}$$

Fundamental tone frequency

$$f_0 = \frac{v}{2\ell} = \frac{500}{2 \times 5} = 50\text{Hz}$$

Second overtone

$$f_2 = \frac{3v}{2\ell} = 150\text{Hz}$$

$$\text{Stress} = \frac{y\Delta\ell}{\ell} = \frac{45 \times 10^{10} \times 12.5 \times 10^{-3}}{5}$$

$$= 1.125 \times 10^7 \text{ W/m}^2$$

9. Ans (A,B,C,D)

$$\text{Velocity of sound wave} = \frac{\omega}{k}$$

$$= \frac{1500\pi}{15\pi} = 100 \text{ m/s}$$

$$\text{Also, } V = \sqrt{\frac{B}{\gamma}} \Rightarrow B = \gamma V^2$$

$$\Rightarrow B = 0.5 \times 100 \times 100$$

$$= 5 \times 10^3 \text{ N/m}^2$$

$$\Delta P_{\max} = 50$$

$$BAk = 50$$

$$A = \frac{50}{5 \times 10^3 \times 15\pi} = \frac{10^{-2}}{15\pi} \text{ m}$$

$$\text{Intensity} = \frac{(\Delta P_{\max})^2}{2\gamma V}$$

$$= \frac{50 \times 50}{2 \times 0.5 \times 100} = 25 \text{ W m}^{-2}$$

10. Ans (A,B)

$$\lambda = \frac{v}{f} = \frac{324}{540} = 60 \text{ cm}$$

$$e = 0.6 r = 3 \text{ cm}$$

$$\ell + e = \frac{\lambda}{4} \text{ or } \frac{3\lambda}{4} \text{ or } \frac{5\lambda}{4}$$

$$\ell = 12 \text{ cm, } 42 \text{ cm, } 72 \text{ cm}$$

$$L = 100 - \ell = 88 \text{ cm, } 58 \text{ cm, } 28 \text{ cm}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (53.33)

$$\text{From } o \text{ to } \frac{L}{2}$$

$$\sqrt{\rho} = \frac{40}{5} \times x \Rightarrow \rho = 64x^2$$

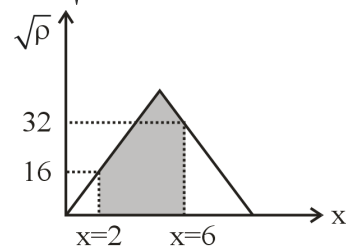
$$\text{Mass} = 2x \int_0^5 64x^2 \times 10^{-5} dx \times 10^3$$

$$= 128 \times 10^{-2} \int_0^5 x^2 dx = \frac{128 \times 125}{3} \times 10^{-2}$$

$$= \frac{160}{3} = 53.33 \text{ g}$$

2. Ans (120.00)

$$v = \sqrt{\frac{T}{\rho A}} \Rightarrow \sqrt{\rho} dx = \sqrt{\frac{T}{A}} dt$$



$$t = \sqrt{\frac{A}{T}} \times \int_{x=2}^{x=6} \sqrt{\rho} dx$$

$$= \sqrt{\frac{10^{-5}}{10}} \times \left[\frac{(40+16)3}{2} + \frac{(40+32) \times 1}{2} \right] \text{ s}$$

$$= 120 \text{ ms}$$

3. Ans (125.60)

$$\pi \left(\frac{\lambda_0 f}{a} \right)^2 = \frac{3.14 \times 16 \times 10^{-4} \times 6.25 \times 10^{-4}}{\frac{1}{4} \times 10^{-6}} \times 10^6 \text{ mm}^2$$

$$= 64 \times 6.25 \times 3.14 \times 10^{-6}$$

$$\text{area} = 1256 \times 10^{-6}$$

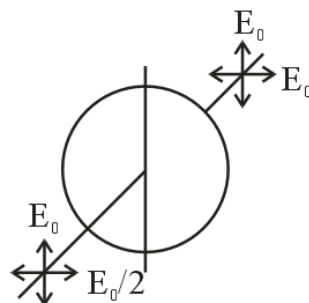
$$x = 125.6$$

4. Ans (1.55 to 1.64)

$$\frac{2 \times 10^{-3}}{1256 \times 10^{-6} \times 10^{-6}} = \frac{2000}{1256} \times 10^6$$

$$= 1.59 \times 10^6$$

5. Ans (75.00)



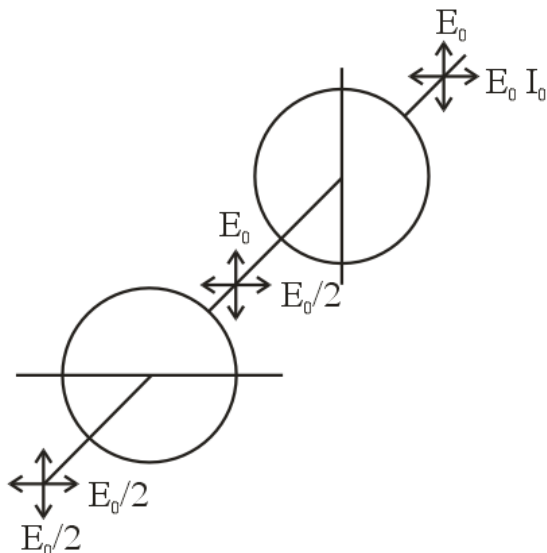
$$\frac{I_1}{I_2} = \frac{(\sqrt{2} E_0)^2}{\left(\sqrt{E_0^2 + \frac{E_0^2}{4}} \right)^2}$$

$$I_2 = \frac{5}{8} \times 120 = 75 \text{ W/m}^2$$

6. Ans (30.00)

$$\frac{I_2}{I_0} = \left(\frac{\sqrt{2} \times \frac{E_0}{2}}{\sqrt{2} E_0} \right)^2$$

$$I_2 = \frac{I_0}{4} = 30 \text{ W/m}^2$$



PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (19)

$$V = \sqrt{\frac{\gamma RT}{M}}, \text{ on mixing O}_2 \quad v' = \frac{v}{2}$$

$$\text{So } M_{eq} = 4M$$

solving for moles

$$n_{H_2} = 4n_{O_2} = N$$

When all oxygen reacts,

$$\frac{N}{2} \text{ moles of H}_2 \text{ and } \frac{N}{2} \text{ moles of H}_2\text{O remains.}$$

$$\text{The effective } \gamma' = \frac{15}{11}, \text{ with H}_2\text{O having } f = 6$$

and

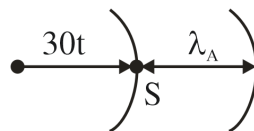
$$\text{effective } M' = 5M$$

$$v' = \sqrt{\frac{15}{77}} \times 1200 \text{ m/s.}$$

$$= \sqrt{\frac{1500}{77}} \times 120 \text{ m/s}$$

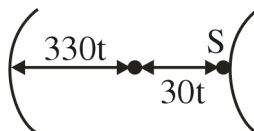
8. Ans (6)

After 't' time



$$330t$$

$$\lambda_A = 330t - 30t = 300t$$



$$30t$$

$$\lambda_B = 330t + 30t = 360t$$

$$\frac{\lambda_A}{\lambda_B} = \frac{300t}{360t} = \frac{10}{12} = \frac{5}{6}$$

$$n = 6$$

9. Ans (8)

$$I_D = \epsilon_0 \frac{d\phi_E}{dt} = \epsilon_0 A \frac{dE}{dt}$$

$$\text{Also } \oint \vec{B} \cdot d\vec{\ell} = B \cdot 2\pi r = \mu_0 I_D$$

$$B = \frac{\mu_0 \epsilon_0 A}{2\pi r} \frac{dE}{dt}$$

$$= \frac{\pi r^2}{2\pi r c^2} \frac{dE}{dt} = \frac{1}{2} \times \frac{r}{c^2} \frac{dE}{dt}$$

$$= \frac{1}{2} \times \frac{0.15}{(3 \times 10^8)^2} \times 10^6$$

$$= 8.33 \times 10^{-13} \text{ T}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B)

(A) $\text{ZnS} \rightarrow$ Wurtzite; Hexagonal
Zinc Blende; Cubic

(B) Correct

(C) Sulphur $\rightarrow$ Rhombic; Orthorhombic

(D) $\text{CaCO}_3 \rightarrow$ Calcite; Trigonal

2. Ans (C)

Solution (A) $V \propto T$ (in kelvin)

$$\therefore \frac{2}{273 + 20} = \frac{V}{273 + 40} \Rightarrow V = 2.14 \text{ l}$$

(B) The volume increases by a factor of

$$\frac{1}{273.15} \text{ of the volume at } 0^\circ\text{C.}$$

3. Ans (A)

$$[\text{H}_2\text{SO}_4] = \frac{2 \times 2}{20} = 0.2\text{M}$$

$$\Rightarrow [\text{H}^+] = 0.4\text{M}$$

$$\therefore \text{pH} = -\log 0.4 = +0.4$$

4. Ans (A)

(A) For single - step, stoichiometry = order

(B) $r = k$, constant at a given temperature

(C) $r = K[\text{C}_2\text{H}_4]$

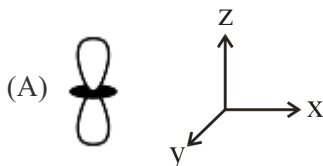
(D) T increases $\Rightarrow$ k increases

$\Rightarrow$ rate increases.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (B,D)



electron density in xy-plane

(B) ψ_{nlm} : Orbital representation.

(C) Exceptional electronic configurations are present.

(D) Angular node = $\ell = 2$, for d-orbital

6. Ans (B,D)

The reaction shifts forward.

(A) Some of the added moles will be consumed but not all.

(B) N_2O_3 increases

(C) N_2O increases

(D) O_2 increases.

7. Ans (A,B,C)

(A) theory

$$(B) \frac{a}{\sqrt{2}}$$

$$(C) \frac{a}{2} = 3 \Rightarrow a = 6 \text{ \AA}$$

$$\text{and } d = \frac{a}{\sqrt{3}} = \frac{6}{\sqrt{3}} = 2\sqrt{3} \text{ \AA}$$

(D) C.N = 6 for OHV.

8. Ans (A,B,C)

$$\lambda = \frac{1}{\sqrt{2\pi\sigma^2 N^*}}; \sigma(\text{He}) < \sigma(\text{Ne})$$

$$\therefore \lambda_{\text{He}} > \lambda_{\text{Ne}}$$

$$\bar{u}_{(\text{He})} > \bar{u}_{(\text{Ne})}$$

$$\text{Also, } z \propto \sigma^2 \bar{u}$$

$$\frac{z_{\text{He}}}{z_{\text{Ne}}} = \left(\frac{140}{154} \right)^2 \sqrt{\frac{20}{4}} > 1$$

$$\Rightarrow z_{\text{He}} > z_{\text{Ne}}$$

$\therefore \lambda$ is independent of temperature for fixed N^* .

$\therefore$ it remains constant

9. Ans (B,D)

$\text{NH}_4\text{Cl} \rightarrow \text{Acidic}$

$\text{KCN} \rightarrow \text{Basic}$

$\text{CH}_3\text{COONH}_4 \rightarrow \text{Neutral}$

$\text{NaHCO}_3 \rightarrow \text{Basic}$

10. Ans (A,C,D)

$$-\int_{A_0}^{A_t} [A] \cdot d[A] = k \cdot \int_0^t dt$$

$$\text{or, } A_0^2 - A_t^2 = 2kt$$

$$\text{Now, } t_{50\%} = \frac{1}{2k} \cdot \left[A_0^2 - \left(\frac{A_0}{2} \right)^2 \right] = \frac{3A_0^2}{8K}$$

$$\text{and } t_{100\%} = \frac{A_0^2}{2K}$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (11.00)

Number of lines = 6 [By each species]

Common transition : $\text{H} : 2 \rightarrow 1$ and $\text{He}^+ : 4 \rightarrow 2$

$\therefore$ Actual spectral lines = $6 + 6 - 1 = 11$.

2. Ans (14.08 to 14.10)

3 times = excitation was upto $n = 3$

$\therefore$ energy incident on atoms

$$= 13.6 \times \left[\frac{1}{1^2} - \frac{1}{3^2} \right] = 12.09 \text{ eV}$$

Now, metal plate consumed 2 eV

$\therefore$ Energy incident on plate = 14.09 eV

$$x = 14.09 - 14.1 \text{ eV}$$

3. **Ans (88.80 to 89.00)**

From the curve,

$T_B = \text{Boyle's temp.} = 300 \text{ K}$

$$\Rightarrow \frac{a}{Rb} = 300k$$

$$T_C = \frac{8a}{27Rb} = \frac{8}{27} \times 300 = 88.89K$$

4. **Ans (0.60)**

$$V_m = \frac{RT}{P} = \frac{0.082 \times 300}{41} = 0.6 \text{ L mol}^{-1}$$

5. **Ans (2.25)**

At 1000 K, open vessel has $P_{\text{air}} = 1 \text{ atm}$.

When sealed and maintained at 1000 K

CaCO_3 would try to attain equilibrium.



The maximum pressure that can be created by the decomposition of the given CaCO_3 is

$$P = \frac{1.5 \times \frac{1}{12} \times 1000}{100} = 1.25 \text{ atm}$$

So, equilibrium can not be attained

and

$$P_{\text{total}} = P_{\text{air}} + P_{\text{CO}_2}$$

$$= 1 + 1.25 = 2.25 \text{ atm}$$

6. **Ans (1.11)**

$\therefore$ the equilibrium was not attained in the original situation

$\therefore$ the added CaCO_3 would dissociate to increase the pressure.

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (3)**

$$K = \frac{A}{\text{Max K}} e^{-E_a/RT}$$

$$3 \times 10^{-3} = A \times 10^{-3} \times 10^{-2}$$

$$A = 300 \text{ min}^{-1}$$

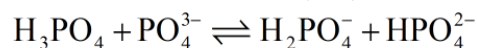
So

$$x = 3$$

8. **Ans (3)**

$$n_{\text{H}_3\text{PO}_4} = 150 \text{ m mol} \quad n_{\text{PO}_4^{3-}} = 50$$

(Acid) (Base)



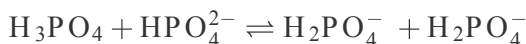
$$\text{initial } 150 \quad 50$$

$$\text{final } 100 \quad - \quad 50 \quad 50$$

$(\text{H}_3\text{PO}_4 \text{ \& } \text{H}_2\text{PO}_4^-)$ and

$(\text{H}_2\text{PO}_4^- \text{ \& } \text{HPO}_4^{2-})$ are conjugates

So, no reaction



$$100 \quad 50$$

$$50 \quad - \quad 50 \quad 50$$

So finally we have 2000 ml solution of H_3PO_4 (50 m mole) and H_2PO_4^- (150 m mole)

a buffer solution

$$\text{pH} = \text{pka} + \log \frac{b}{a}$$

$$\text{pH} = 2.5 + \log 3 = 3$$

9. **Ans (150)**

$$\frac{r_x}{r_{\text{O}_2}} = \frac{4}{5} = \sqrt{\frac{32}{X}}$$

$$x = 50$$

$$PM = ZdT$$

$$Z = \frac{1 \times 50}{\frac{1}{12} \times 0.8 \times 500} = 1.5$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

Use the identity

$$\left(\cos \theta - \cos \frac{\pi}{n} \right) \left(\cos \theta - \cos \frac{2\pi}{n} \right) \dots$$

$$\left(\cos \theta - \cos \frac{(n-1)\pi}{n} \right)$$

$$= \frac{\sin n\theta}{2^{n-1} \sin \theta}$$

$$\text{Now } \int \frac{\sin 7\theta}{\prod_{k=1}^6 \left(\cos \theta - \cos \frac{k\pi}{7} \right)} d\theta$$

$$= \int \frac{\sin 7\theta}{\left(\frac{\sin 7\theta}{2^6 \sin \theta} \right)} d\theta = \int 2^6 \sin \theta d\theta$$

$$= -2^6 \cos \theta + C$$

2. Ans (D)

$$I = \int_0^{\infty} \frac{\sin\left(x - \frac{1}{x}\right)}{x} dx$$

$$\text{put } x = \frac{1}{t}$$

$$dx = -\frac{1}{t^2} dt$$

$$I = \int_{\infty}^0 \frac{\sin\left(\frac{1}{t} - t\right) \left(\frac{-1}{t^2}\right) dt}{1/t}$$

$$\Rightarrow I = 0$$

3. Ans (A)

$$\text{Let } x + y = A \text{ and } x - y = B$$

$$Bf(A) - Af(B) = (A^2 - B^2) AB$$

$$\frac{f(A)}{A} = A^2 \quad \frac{f(B)}{B} = B^2$$

$$f(A) = A^3 \quad f(B) = B^3$$

$$\Rightarrow f(x) = x^3$$

$$\text{curves are } y^2 + x = 0 \text{ and } x^2 + y^2 = 12$$

Required area

$$= 2 \int_{-2\sqrt{3}}^{-3} \sqrt{12 - x^2} dx + 2 \int_{-3}^0 \sqrt{-x} dx$$

$$= 2 \int_3^{2\sqrt{3}} \sqrt{12 - x^2} dx + 2 \int_0^3 \sqrt{x} dx$$

(replace $-x$ in place of x)

$$= 2\pi + \sqrt{3} \text{ sq. unit}$$

4. Ans (B)

$$2x + 2yy' = e^{\left(\frac{x^2+y^2}{x}\right)} + \frac{x^2+y^2}{x}$$

$$\text{Put } \frac{x^2+y^2}{x} = v$$

$$\Rightarrow x^2 + y^2 = vx$$

$$2x + 2yy' = v + x \left(\frac{dv}{dx}\right)$$

from the given differential equation, we get

$$v + x \frac{dv}{dx} = e^v + v$$

$$e^{-v} dv = \frac{dx}{x}$$

$$-e^{-v} = \log cx$$

$$cx = e^{-e^{-v}} = \frac{1}{e^{e^{-\left(\frac{x^2+y^2}{x}\right)}}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,D)

$$\text{Let } \frac{\sqrt[4]{x^{10} + x^8 + 1}}{x} = t$$

$$x \left(\frac{10x^9 + 8x^7}{4(x^{10} + x^8 + 1)^{\frac{3}{4}}} \right) - (x^{10} + x^8 + 1)^{\frac{1}{4}} \Rightarrow \frac{x^2}{4(x^{10} + x^8 + 1)^{\frac{3}{4}}} dx = dt$$

$$\Rightarrow \frac{10x^{10} + 8x^8 - 4(x^{10} + x^8 + 1)}{4x^2(x^{10} + x^8 + 1)^{3/4}} dx = dt$$

$$\Rightarrow \frac{6x^{10} + 4x^8 - 4}{4x^2(x^{10} + x^8 + 1)^{3/4}} dx = dt$$

$$\Rightarrow \frac{(3x^{10} + 2x^8 - 2)dx}{2x^2 \left(\frac{\sqrt[4]{x^{10} + x^8 + 1}}{x} \right)^3 \cdot x^3} = dt$$

$$\Rightarrow \frac{(3x^{10} + 2x^8 - 2) dx}{2x^5 t^3} = dt$$

$$dx = \frac{2x^5 t^3 dt}{3x^{10} + 2x^8 - 2}$$

$$\text{Now, } I = \int 2x^5 t^3 \frac{\sqrt[4]{x^{10} + x^8 + 1}}{x \cdot x^5} dt$$

$$= 2 \int t^4 dt = \frac{2}{5} \cdot t^5 + C$$

$$= \frac{2}{5} \left(\frac{(x^{10} + x^8 + 1)^{\frac{5}{4}}}{x^5} \right) + C$$

6. Ans (A,B)

Case-I;

$$x \in [0, 4]$$

$$f(x) = \int_0^x (x-t)dt + \int_x^4 (t-x)dt = x^2 - 4x + 8$$

Case-II; ($x > 4$)

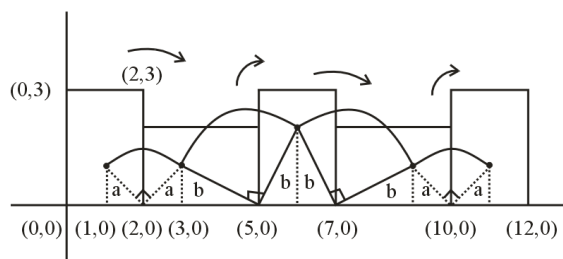
$$f(x) = \int_0^4 (x-t)dt = 4x - 8$$

Case-III; ($x < 0$)

$$f(x) = \int_0^4 (t-x)dt = 8 - 4x$$

Now draw graphs and check.

7. Ans (C,D)



Req. area = (Ar. of 4 Δ 's of area $\frac{1}{2}$)

+ (Ar. of 4 Δ 's with area 1)

+ (Ar. of 2 quarter circles of area $\frac{\pi}{4}(\sqrt{2})^2$)

+ (Ar. of 2 quarter circles of area $\frac{\pi}{4}(\sqrt{5})^2$)

$$= 6 + \frac{7\pi}{2}$$

8. Ans (A,D)

$$1 < \frac{\pi}{2} < 2$$

$$x \in (0, 1)$$

then

$$x^{\frac{\pi}{2}} < x^1$$

$$1 + x^{\frac{\pi}{2}} < 1 + x$$

$$\frac{1}{1 + x^{\frac{\pi}{2}}} > \frac{1}{1 + x}$$

$$\int_0^1 \frac{1}{1 + x^{\frac{\pi}{2}}} dx > \int_0^1 \frac{1}{1 + x} dx = \ln 2$$

$$x \in (0, 1)$$

$$x^{\frac{\pi}{2}} > x^2$$

$$1 + x^{\frac{\pi}{2}} > 1 + x^2$$

$$\frac{1}{1 + x^{\frac{\pi}{2}}} < \frac{1}{1 + x^2}$$

$$\int_0^1 \frac{1}{1 + x^{\frac{\pi}{2}}} dx < \int_0^1 \frac{1}{1 + x^2} dx < \frac{\pi}{4}$$

9. Ans (A,C)

$$A = \int_a^c (f(c) - f(x)) dx + \int_c^b (f(x) - f(c)) dx$$

$$= f(c)(c - a) - f(c)(b - c) - \int_a^c f(x) dx + \int_b^c f(x) dx$$

Let

$$g(c) = f(c)(2c - a - b) - \int_a^c f(x) dx - \int_b^c f(x) dx$$

$$g'(c) = f'(c)(2c - a - b)$$

area is min when $c = \frac{a+b}{2}$ and max at $c = a$ or $c = b$ for options (C) & (D)

$$\text{Area is min at } x = \frac{0+2}{2} = 1$$

$$\text{So } f(1) = 0 \quad \frac{1}{3} - 1 + a = 0$$

$$a = \frac{2}{3}$$

10. Ans (B)

$$\frac{dy}{dx} = \frac{3x^2y^4 + 2xy}{x^2 - 2x^3y^3}$$

$$x^2 dy - 2x^3y^3 dy = 3x^2y^4 dx + 2xy dx$$

$$x^2 dy - 2xy dx = 3x^2y^4 dx + 2x^3y^3 dy$$

$$= y^2 [3x^2y^2 dx + 2x^3y dy]$$

$$-[2xy dx - x^2 dy] = y^2 d(x^3y^2)$$

$$-\left[\frac{2xy dx - x^2 dy}{y^2}\right] = d(x^3y^2)$$

$$-d\left(\frac{x^2}{y}\right) = d(x^3y^2)$$

$$-\frac{x^2}{y} = x^3y^2 + C$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (1.00)

$$\text{At } x = 1 \quad y = 0$$

$$\frac{dy}{dx} = 2x(x^4)^2 - (x^2)^2$$

$$\frac{dy}{dx} \Big|_{x=1} = 2 - 1 = 1$$

$$\therefore \text{Tangent } y - 0 = 1(x - 1) \Rightarrow y = x - 1$$

2. Ans (0.00)

$$F(x) = \int_1^x e^{\frac{t^2}{2}} (1 - t^2) dt$$

$$\Rightarrow F'(x) = \left(e^{x^2/2} (1 - x^2) \right)^2$$

$$\Rightarrow F'(1) = 0$$

3. Ans (4.00)

$$g(x+y) = e^y g(x) + e^x g(y)$$

$$\text{Put } x = 0 = y$$

$$g(0) = 0$$

$$g'(0) = 2$$

$$\text{so } \lim_{h \rightarrow 0} \frac{g(h)}{h} = 2$$

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{e^h g(x) + e^x g(h) - g(x)}{h}$$

$$g'(x) = g(x) \lim_{h \rightarrow 0} \frac{e^h - 1}{h} + e^x \lim_{h \rightarrow 0} \frac{g(h)}{h}$$

$$g'(x) = g(x) + 2e^x$$

$$\text{If } g(x) = y \text{ then } \frac{dy}{dx} = y + 2e^x$$

$$\frac{dy}{dx} - y = 2e^x, \text{ IF } = e^{\int -1 dx} = e^{-x}$$

$$ye^{-x} = \int 2e^x e^{-x} dx + C$$

$$ye^{-x} = 2x + C, g(0) = 0$$

$$0 = C$$

$$\text{So } y = 2xe^x$$

4. Ans (4.00)

$$g(x) = 2xe^x$$

$$g'(x) = 2e^x + 2xe^x = 0$$

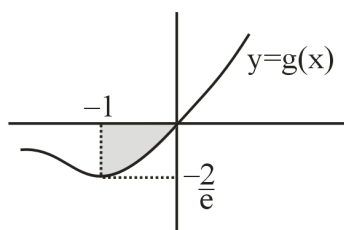
$$2e^x(x+1) = 0$$

$$x = -1$$

$$g'(x) > 0 \text{ for } x > -1 \text{ and}$$

$$g'(x) < 0 \text{ for } x < -1$$

$$\text{for } x < 0, g(x) < 0, x > 0, g(x) > 0$$



$$\text{So required area} = \left| 2 \int_{-1}^0 xe^x dx \right| = \left(2 - \frac{4}{e} \right)$$

5. Ans (7.00)

$$f(x) = \{x\} \sin(\pi x)$$

$\{x\}$ is periodic with period 1.

$\sin(\pi x)$ is periodic with period $\frac{2\pi}{\pi} = 2$

LCM of 1 & 2 is period of $f(x)$.

$f(x)$ is periodic with period '2'.

Let area bounded by the graph and x-axis between 0 & 1 is 'A' then

$$A = \int_0^1 \{x\} \sin(\pi x) dx = \int_0^1 x \sin(\pi x) dx$$

$$= \left[-x \frac{\cos \pi x}{\pi} + \int \frac{\cos \pi x}{\pi} dx \right]_0^1 \quad (\text{Integration by parts})$$

by parts)

$$= \left[-x \frac{\cos \pi x}{\pi} + \frac{\sin \pi x}{\pi^2} \right]_0^1$$

$$= \left[\frac{1}{\pi} + 0 - 0 + 0 \right] = \frac{1}{\pi}$$

(1) Require area between $x = -3$ and $x = b$

$$= \int_{-3}^b |\{x\} \sin(\pi x)| dx = \frac{10}{\pi}$$

$$\Rightarrow (b - (-3)) \int_0^1 \{x\} \sin(\pi x) dx = \frac{10}{\pi}$$

$$\Rightarrow (b + 3) \frac{1}{\pi} = \frac{10}{\pi} \Rightarrow b = 7$$

($\because$ between two consecutive integer it cover $\frac{1}{\pi}$ unit area)

6. Ans (0.00)

Area bounded by the curve, x-axis, $x = -\frac{1}{2}$

and $x = \frac{5}{2}$ will be equal to :

$$= 3 \int_0^1 \{x\} \sin(\pi x) dx = \frac{3}{\pi}$$

$$\text{because } \frac{5}{2} - \left(\frac{-1}{2} \right) = 3$$

$$\left[\frac{3}{\pi} \right] = 0$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (1)**

$$x^2 \int_0^1 \frac{dt}{t^2 + 2t \sin x + 1} - x \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{t^2 \tan 3t}{t^2 + 5} dt - \frac{1}{4} = 0$$

$$x^2 I_1 - x I_2 - \frac{1}{4} = 0$$

$$I_1 = \int_0^1 \frac{dt}{t^2 + 2t \sin x + 1}$$

$$= \int_0^1 \frac{dt}{t^2 + 2t \sin x + \sin^2 x + \cos^2 x}$$

$$= \int_0^1 \frac{dt}{(t + \sin x)^2 + \cos^2 x}$$

$$= \frac{1}{\cos x} \tan^{-1} \left(\frac{t + \sin x}{\cos x} \right) \Big|_0^1$$

$$= \frac{1}{\cos x} \left[\tan^{-1} \left(\frac{1 + \sin x}{\cos x} \right) - \tan^{-1} \left(\frac{\sin x}{\cos x} \right) \right]$$

$$= \frac{1}{\cos x} \left(\tan^{-1} \left(\frac{\sin \frac{x}{2} + \cos \frac{x}{2}}{\cos \frac{x}{2} - \sin \frac{x}{2}} \right) - \tan^{-1} (\tan x) \right)$$

$$= \frac{1}{\cos x} \left(\tan^{-1} \left(\frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right) - x \right)$$

$$= \frac{1}{\cos x} \left(\tan^{-1} \left(\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right) - x \right)$$

$$I_1 = \frac{1}{\cos x} \left(\frac{\pi}{4} + \frac{x}{2} - x \right)$$

$$= \frac{1}{\cos x} \left(\frac{\pi}{4} - \frac{x}{2} \right) \quad \dots(1)$$

$$I_2 = \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{t^2 \tan 3t}{t^2 + 5} dt = 0 \text{ as function is odd}$$

so equation is

$$x^2 \left(\frac{1}{\cos x} \left(\frac{\pi}{4} - \frac{x}{2} \right) \right) - \frac{1}{4} = 0$$

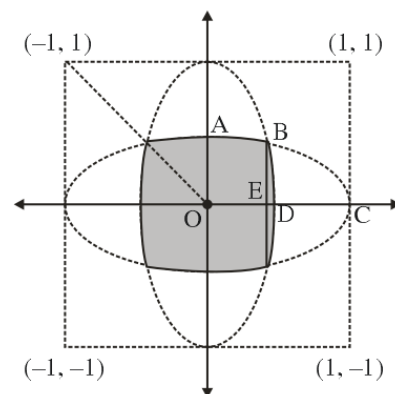
$$x^2 (\pi - 2x) = \cos x$$

This equation has one solution

8. **Ans (0)**

Let $P(x, y)$

$$OP = \sqrt{x^2 + y^2}$$



$$\sqrt{x^2 + y^2} \leq |1 - x| \quad \dots(1)$$

$$\sqrt{x^2 + y^2} \leq |1 + x| \quad \dots(2)$$

$$\sqrt{x^2 + y^2} \leq |1 - y| \quad \dots(3)$$

$$\sqrt{x^2 + y^2} \leq |1 + y| \quad \dots(4)$$

Solving equation (1)

$$x^2 + y^2 \leq 1 + x^2 - 2x$$

$$y^2 \leq 1 - 2x, \text{ inside the parabola } y^2 = 1 - 2x$$

$$S = 4 \text{ (area OABDO)}$$

$$= 4 \text{ (area of OABEO + Area BDEB)}$$

Point B is point of intersection of

$$y^2 = 1 - 2x \text{ and } x^2 = 1 - 2y$$

$$y^2 - x^2 = -2(x - y)$$

$$x = y \text{ or } x + y = 2$$

$$\text{If } y = x \text{ then } x^2 = 1 - 2x$$

$$-1 - \sqrt{2} \text{ is not possible as it is outside}$$

$$\text{so } x = \sqrt{2} - 1$$

There is no real root when $x + y = 2$

$$\text{so point B} = (\sqrt{2} - 1, \sqrt{2} - 1)$$

Area

$$S = 4 \left\{ \int_0^{\sqrt{2}-1} \frac{1-x^2}{2} + \int_{\sqrt{2}-1}^{\frac{1}{2}} \sqrt{1-2x} dx \right\}$$

$$= \frac{1}{3} (16\sqrt{2} - 20) = \frac{1}{3} (16 \times 1.4 - 20)$$

$$= \frac{1}{3} (22.4 - 20) = \frac{2.4}{3} = 0.80$$

Alternate Method :

As the shaded figure lies in square of size 1×1
so its G.I.F. values is zero.

9. Ans (2)

$$x \cos \frac{y}{x} d(xy) = y \sin \frac{y}{x} d\left(\frac{y}{x}\right) \cdot x^2$$

$$\cos\left(\frac{y}{x}\right) d(xy) = xy \sin\left(\frac{y}{x}\right) d\left(\frac{y}{x}\right)$$

$$\int \frac{d(xy)}{xy} = \int \tan\left(\frac{y}{x}\right) d\left(\frac{y}{x}\right)$$

$$\ln(xy) = \ln \left| \sec \frac{y}{x} \right| + \ln C$$

$$xy = \sec\left(\frac{y}{x}\right) C$$

$$\cos\left(\frac{y}{x}\right) xy = C$$

$$\text{Put } x = 1, y(1) = 2\pi$$

$$2\pi \cos(2\pi) = C$$

$$C = 2\pi$$

$$\cos\left(\frac{y}{x}\right) xy = 2\pi$$

$$\text{put } x = 4$$

$$\cos\left(\frac{y(4)}{4}\right) \cdot 4 \cdot y(4) = 2\pi$$

$$y(4) \cos\left(\frac{y(4)}{4}\right) = \frac{\pi}{2}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PART TEST

13-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C,D	B,D	A,B,D	B,C,D	B,C,D	C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	A	D	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.18 to 2.21	602.00	954.54 to 954.55	4.71	0.80	22.50
SECTION-II (ii)	Q.	7	8	9			
	A.	8	7	2750			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C,D	C,D	A,B	C	A,C,D	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	A	A	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.40	1.50	53.33	47.14	78.50 to 78.60	0.74
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	160			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C	A,D	A,C,D	A,C,D	C,D	A,B
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	D	C	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.50	4.50	2.00	8.00	261.00	14.00
SECTION-II (ii)	Q.	7	8	9			
	A.	12	2	1			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,C,D)

$$I = \frac{P}{A} = \frac{1.77 \times 10^3 \text{ W}}{0.2 \text{ m}^2} = 8.85 \times 10^3 \text{ W/m}^2$$

Peak intensity is double average intensity

$$I = \frac{1}{2} \epsilon_0 E_0^2 C$$

$$E_0^2 = \frac{2I}{\epsilon_0 C} = \frac{2 \times 8.85 \times 10^3}{8.85 \times 10^{-12} \times 3 \times 10^8} = 6.66 \times 10^6 \text{ N}^2/\text{C}^2$$

$$\left(\frac{1}{2\mu_0} B_0^2 \right) = \frac{I}{C} = \frac{8.85 \times 10^3}{3 \times 10^8} = 2.95 \times 10^{-5} \text{ J/m}^3$$

2. Ans (B,D)

Theory based questions

3. Ans (A,B,D)

$$y \text{ (1st minima)} = 225 \text{ W}$$

$$W \sin \theta = \lambda$$

$$\rightarrow \sin \theta = \frac{\lambda}{W} = \tan \theta = \frac{225}{18000} = \frac{1}{80}$$

For y(2nd minima),

$$W \sin \theta = 2\lambda \rightarrow \tan \theta = \frac{1}{40}$$

$$I = I_0 \left[\frac{\sin\left(\frac{\phi}{2}\right)}{\left(\frac{\phi}{2}\right)} \right]^2,$$

$$\Delta x = W \sin \theta = \frac{W}{480} = \frac{\lambda}{6} \rightarrow \phi = \frac{2\pi}{\lambda} (\Delta x)$$

$$\phi = \frac{\pi}{3}$$

4. Ans (B,C,D)

Theory based questions

5. Ans (B,C,D)

$$L_1 = 10 \log_{10} \frac{I_1}{I_0}$$

$$L_2 = 10 \log_{10} \frac{I_2}{I_0}$$

$$\frac{I_1}{I_2} = 10^{0.4} = (10^{0.2})^2 = 2.56$$

$$\frac{I_1}{I_2} = \left(\frac{r_2}{r_1} \right)^2, \quad \frac{r_2}{r_1} = 10^{0.2} = 1.6$$

$$r_2 = 1.6 r_1 \quad \text{also} \quad r_2 - r_1 = 3$$

$$1.6r_1 - r_1 = 3 \Rightarrow 0.6r_1 = 3$$

$$\Rightarrow r_1 = 5 \text{ m}$$

$$r_2 = 1.6 \times 5 = 8 \text{ m.}$$

6. Ans (C)

Acceleration of the pulse remains constant in all the three strings.

$$\frac{\lambda_B}{\lambda_C} = \frac{v_B}{v_C} = \sqrt{\frac{T_B}{T_C}} \times \sqrt{\frac{\mu_C}{\mu_B}} = \sqrt{\frac{5}{3}} \times \sqrt{\frac{3}{1}} = \sqrt{5} : 1$$

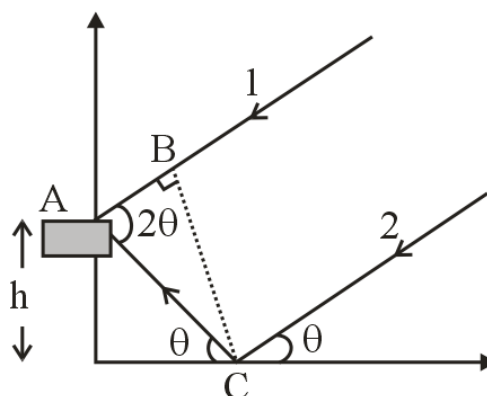
PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (C)

The signal detected by the detector A, is a result of interference of two rays the ray 1 incident directly from the star and ray 2, reflected from sea surface.

The phase of second ray is shifted by π due to the reflection by a denser medium.



Therefore the path difference between the two rays is :

$$\Delta x = AC + \frac{\lambda}{2} - AB = \frac{h}{\sin \theta} + \frac{\lambda}{2} - \frac{h}{\sin \theta} \cos 2\theta$$

$$= \frac{\lambda}{2} + \frac{h}{\sin \theta} (1 - \cos 2\theta) = \frac{\lambda}{2} + 2h \sin \theta$$

Condition for 2nd maxima is :

$$\frac{\lambda}{2} + 2h \sin \theta = 2\lambda$$

$$\sin \theta = \frac{3\lambda}{4h} = \frac{3}{4} \times \frac{21\text{cm}}{210\text{cm}} = 0.075$$

So N = 75

8. **Ans (A)**

$$E_{\max} = E_i + E_r = E_i + E_i \left(\frac{n - \cos i}{n + \cos i} \right)$$

$$\text{Since } i = \frac{\pi}{2} - \theta$$

$$E_{\max} = E_i \left(1 - \frac{n - \sin \theta_{\max}}{n + \sin \theta_{\max}} \right)$$

$$= E_i \frac{2n}{n + \sin \theta_{\max}}$$

$$\frac{I_{\text{1st maxima}}}{I_{\text{next minima}}} = \left(\frac{n^2}{\sin^2 \alpha_{\min}} \right) \left(\frac{n + \sin \alpha_{\min}}{n + \sin \alpha_{\max}} \right)^2$$

For 1st maxima

$$\rightarrow \frac{\lambda}{2} + 2h \sin \theta = \lambda$$

$$\rightarrow \sin \theta = \frac{\lambda}{4h} = \frac{21}{4 \times 210}$$

$$\rightarrow \sin \theta_{\max} = 0.025$$

For next minima

$$\rightarrow \frac{\lambda}{2} + 2h \sin \theta = \frac{3}{2}\lambda$$

$$\rightarrow \sin \theta = \frac{\lambda}{2h}$$

$$\rightarrow \sin \theta_{\min} = 0.05$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{9}{0.05} \right)^2 \times \left(\frac{9 + 0.05}{9 + 0.025} \right)^2$$

Ans. 3.25 - 3.26

9. **Ans (D)**

The frequencies are in the ratio of 5 : 7 : 9.

Fixed at one end & free at other.

The fundamental frequency is the H.C.F. of 175, 245 and 315, so $f_0 = 35$ Hz

$$175 \text{ Hz is } 5^{\text{th}} \text{ Harmonic} \rightarrow \lambda_1 = \frac{2L}{5}$$

$$245 \text{ Hz is } 7^{\text{th}} \text{ Harmonic} \rightarrow \lambda_2 = \frac{2L}{7}$$

$$315 \text{ Hz is } 9^{\text{th}} \text{ Harmonic} \rightarrow \lambda_3 = \frac{2L}{9}$$

$$\alpha + \beta + \gamma = 11.68 \text{ m.}$$

10. **Ans (A)**

With two nodes and antinodes, it is 3rd harmonic

$$\lambda = \frac{4L}{3} = 17.16 \text{ m}$$

$$v = \lambda f = 600.6 \text{ m/s}$$

$$T = \mu v^2 = 360.72 \text{ N}$$

PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (2.18 to 2.21)**

$$f_h = \left(\frac{v}{v - v_s} \right) f, f_L = \left(\frac{v}{v + v_s} \right) f$$

$$f_h - f_L = \left[\frac{v}{v - v_s} - \frac{v}{v + v_s} \right] f$$

$$= \left[\frac{1}{1 - \frac{v_s}{v}} - \frac{1}{1 + \frac{v_s}{v}} \right] f$$

$$= \left[\left(1 - \frac{v_s}{v} \right)^{-1} - \left(1 + \frac{v_s}{v} \right)^{-1} \right] f$$

Since difference is small $\frac{v_s}{v} \ll 1$

So,

$$f_h - f_L = \left[\left(1 + \frac{v_s}{v} \right) - \left(1 - \frac{v_s}{v} \right) \right] f = \frac{2v_s}{v} f$$

$$\rightarrow 2 = \frac{2v_s}{330} \times 600$$

$$\rightarrow v_s = \frac{330}{600} = \frac{11}{20} \text{ m/s.}$$

$$v_s = A\omega = 2\pi f A = 8 \times \frac{22}{7} \times A$$

$$A = \frac{7}{320} \text{ m} = 2.19$$

2. **Ans (602.00)**

$\omega = \sqrt{\frac{k}{m}}$, Half length spring has double k and

mass is also halved, So $\omega' = 2\omega$,

$$v_s' = 2v_s \text{ previous } \frac{v_s}{v} = \frac{1}{600}$$

New

$$\frac{v_s'}{v} = \frac{2}{600},$$

$$f' = \left(1 + \frac{v_s'}{v} \right) 600 = 602 \text{ Hz}$$

3. **Ans (954.54 to 954.55)**

$$I = \frac{P_0}{4\pi r^2}$$

$$P = IA \cos \theta$$

$$P = \frac{P_0}{4\pi r^2} A \cos \theta$$

$$= \frac{4 \times 10^{26} \times 0.75 \times 0.9}{4 \times \frac{22}{7} \times 2.25 \times 10^{22}}$$

$$= \frac{2.1}{22} \times 10^4$$

$$= 954.54$$

4. **Ans (4.71)**

Intensity = Total energy density $\times$ speed of radiation

$$U = \frac{I}{C} = \frac{P_0}{4\pi r^2 C}$$

$$= \frac{4.0 \times 10^{26}}{4 \times \frac{22}{7} \times (1.5 \times 10^{11})^2 \times 3 \times 10^8}$$

$$= \frac{7 \times 10^4}{22 \times 2.25 \times 3 \times 10^8}$$

$$= \frac{700}{148.5} \times 10^{-6}$$

$$= 4.71 \times 10^{-6} \text{ Jm}^{-3}$$

5. **Ans (0.80)**

For second minima $a \sin \theta = 2\lambda$

$$\sin \theta = \frac{2\lambda}{a} = \frac{2 \times 6 \times 10^{-7}}{0.6 \times 10^{-3}}$$

$$\sin \theta = 2 \times 10^{-4} = \tan \theta$$

The diffraction pattern is forming on the screen at a distance of 20 cm distance y from central maxima on the screen.

$$\text{so, } \tan \theta = \frac{y}{20\text{cm}} = 2 \times 10^{-4}$$

$$y = 40 \times 10^{-4} \text{ cm.}$$

Separation between two minimas

$$= 2y = 80 \times 10^{-4} \text{ cm.}$$

$$= 0.08 \text{ mm.}$$

6. **Ans (22.50)**

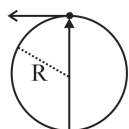
At central maxima all the waves originating from the plane wave front superpose constructively. Let the resultant amplitude be A



At 1st maxima path difference between one end of wave front and the other is $\frac{3\lambda}{2}$, i.e. phase difference of 3π

So resultant amplitude is $2R$

$$3\pi R = A \Rightarrow R = \frac{A}{3\pi} \quad A' = \frac{2A}{3\pi}$$



Ratio of intensity

$$= \frac{A^2}{\left(\frac{2A}{3\pi}\right)^2} = \frac{9\pi^2}{4} = \frac{90}{4} = 22.5$$

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (8)**

Path difference at source

$$= d \sin \theta = 0.1 \times 0.03\text{mm} = 3 \times 10^{-6} \text{ m}$$

$$= 5 \times 0.6 \times 10^{-6} \text{ m}$$

$$= 5 \lambda$$

Since 3 minimas and two maximas have gone past O

So new path difference due to thin film $\frac{8\lambda}{3}$

$$\frac{8}{3}\lambda = (\mu - 1) t$$

$$t = \frac{8}{3} \frac{\lambda}{(\mu - 1)} = \frac{8 \times 0.6\mu\text{m}}{3 \times (1.2 - 1)} = 8\mu\text{m}$$

8. **Ans (7)**

Intensity after 1st polaroid is $\frac{I_0}{2}$

for each subsequent polariser $I = \frac{I_0}{2} (\cos^2 \theta)^n$

Where n is the number of subsequent polarisers.

$$I = \frac{I_0}{2} \left(\frac{1}{2}\right)^n = \frac{I_0}{2^{n+1}} < \frac{I_0}{100}$$

Minimum value of n satisfying this is $n = 6$

So total polariser is $n + 1 = 7$

9. **Ans (2750)**

When frequency of sound matches with frequency of natural vibration of air column, resonance happens Which causes the amplitude to become large. So resonant frequencies will be heard best

$$f_n = (2n - 1) \frac{v}{4\ell}$$

$$f_1 = \frac{v}{4\ell} = \frac{330 \text{ ms}^{-1}}{4 \times 0.03} = 2750$$

$$f_2 = 3 \frac{v}{4L} = 8250$$

So in the given range the best heard frequency will be 2750 Hz.

PART-2 : CHEMISTRY

SECTION-I (i)

- Ans (A,C,D)**
 $\rho_{\text{exp.}} < \rho_{\text{theo.}}$
- Ans (C,D)**
 $d = \frac{PM}{RT}$
- Ans (A,B)**
 Factual
- Ans (C)**
 NaOH reacts with CO_2 & H_2O present in air.
- Ans (A,C,D)**
 Apply L.C.P.
- Ans (A,C)**
 $\lambda = \frac{h}{mv}$;

PART-2 : CHEMISTRY

SECTION-I (ii)

- Ans (D)**
 $\Psi(r, \theta, \phi) \propto r^\ell$
 $\therefore \ell = 1$
 No. of radial node = 3
 Radial node at :
 $(1 - \sigma)(12 - 8\sigma + \sigma^2) = 0$
 $\therefore n = 5$
 $\therefore (n + \ell) = 6$
- Ans (A)**
 Angular Node at :
 $\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2} \Rightarrow xy \text{ plane}$
 $\therefore$ Orbital is p_z
 $\therefore m_\ell = 0$
- Ans (B)**
 For first drop of $\text{C}(\ell)$ formed, pressure due to vapours of C should be 0.6 atm. For it, moles of C formed should be $\frac{0.6 \times 16.4}{0.082 \times 300} = 0.4$.

PART-2 : CHEMISTRY

SECTION-II (i)

- Ans (0.40)**
 $3\text{A}_{2(g)} \rightleftharpoons \text{A}_{6(g)}$
 $K_p = 1.6 = \frac{0.2}{p_{\text{A}_2}^3}$
 $\Rightarrow p_{\text{A}_2} = 0.5 \text{ atm}$
 Let the partial pressure of A_2 and C before the start of reaction be a and a/2 atm respectively.
 $\text{A}_{2(g)} + \text{C}_{(g)} \rightleftharpoons \text{A}_2\text{C}_{(g)}$
 If x moles of A_2 reacts to form x moles of A_2C .
 $1.4 = 0.5 + 0.2 + p_{\text{A}_2\text{C}} + p_{\text{C}}$
 $\Rightarrow p_{\text{A}_2\text{C}} + p_{\text{C}} = 0.7$
 $a/2 - x + x = 0.7$
 $a = 1.4 \text{ atm}$
 Now amount of A_2 Consumed in first reaction
 $= 3 \times 0.2 = 0.6 \text{ atm}$
- Ans (1.50)**
 Amount of A_2 consumed in second reaction
 $= 1.4 - 3 \times 0.2 - 0.5 = 0.3 \text{ atm}$
 $\Rightarrow p_{\text{A}_2\text{C}} = 0.3 \text{ atm}$
 $p_{\text{C}} = 0.7 - 0.3 = 0.4 \text{ atm}$
 $K_p = \frac{0.3}{0.4 \times 0.5} = 1.5 \text{ atm}^{-1}$
- Ans (53.33)**
 $P = \frac{nRT}{V}$
 $= \frac{1.00 \times 0.08 \times 400}{0.60} = 53.33 \text{ atm.}$
- Ans (47.14)**
 $\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$
 $P = 47.14$
- Ans (78.50 to 78.60)**
 $\eta = \frac{2 \times \pi R^2}{a^2}$
 $a\sqrt{2} = 4R$
 $\eta = \frac{2 \times \pi R^2}{4 \times 4 \times R^2 \times \frac{1}{2}} = \frac{\pi}{4}$

6. Ans (0.74)

$$FCC ; \eta = 74\%$$

Ans : 74

PART-2 : CHEMISTRY

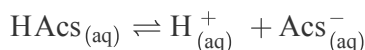
SECTION-II (ii)

7. Ans (3)

$$t_{\frac{1}{2}} \propto a^{1-n}$$

$$n = 3$$

8. Ans (3)



c

$$c(1-\alpha) \quad c\alpha \quad c\alpha$$

$$K_a = \frac{c\alpha^2}{1-\alpha}$$

$$c = \frac{0.325}{0.5} = 3.6 \times 10^{-3}$$

$$\Rightarrow 10\alpha^2 + \alpha - 1 = 0$$

$$\alpha = 0.27$$

$$[H^+] = 0.27 \times 3.6 \times 10^{-3}$$

$$pH = 3.0$$

9. Ans (160)

As we know

$$pV = nRT$$

$$p = \frac{n}{V} RT$$

$$\Rightarrow p = \frac{\left(\frac{w}{M}\right) RT}{V} \quad \therefore n = \frac{w}{M}$$

$$\Rightarrow p = \frac{w \cdot RT}{V \cdot M} = \frac{w}{V} \times \frac{RT}{M}$$

$$\Rightarrow p = d \cdot \frac{RT}{M} \quad \left(\because \frac{w}{V} = d \right)$$

$$\Rightarrow d = \frac{pM}{RT}$$

Multiply both sides by p

$$\Rightarrow pd = p \times \frac{pM}{RT} = \frac{p^2 M}{RT} \quad \dots(i)$$

On differentiating Eq. (i) w.r.t. p, we will get

$$\frac{d(p \cdot d)}{dp} = \frac{d}{dp} \left(\frac{p^2 M}{RT} \right) = \frac{2pM}{RT}$$

$$5 = \frac{2pM}{RT}$$

$$T = \frac{2pM}{5R} = \frac{2 \times 8.21 \times 4}{5 \times 0.0821} \quad [\text{for He, } M = 4]$$

$$T = 160 \text{ K}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C)

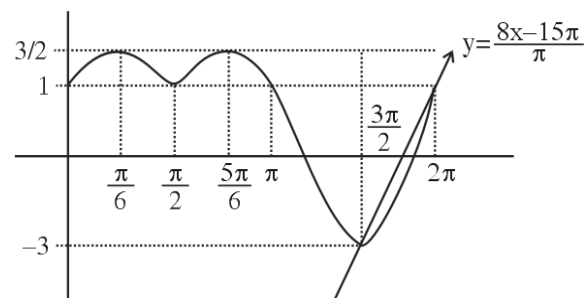
$$y = 2 \sin x + \cos 2x$$

$$y' = 2 \cos x - 2 \sin 2x$$

$$\frac{dy}{dx} = 2 \cos x (1 - \sin 2x)$$

$$\frac{dy}{dx} = 0 \Rightarrow x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

$$\frac{dy}{dx} \Rightarrow \begin{array}{cccccc} + & - & + & - & + \\ \frac{\pi}{6} & \frac{\pi}{2} & \frac{5\pi}{6} & \frac{3\pi}{2} \end{array}$$



$$A = \int_0^{\frac{3\pi}{2}} (2 \sin x + \cos 2x) dx + \int_{\frac{3\pi}{2}}^{2\pi} \left(\frac{8x - 15\pi}{\pi} \right) dx$$

$$A = \left[-2 \cos x + \frac{\sin 2x}{2} \right]_0^{\frac{3\pi}{2}} + \frac{1}{\pi} (4x^2 - 15\pi x) \Big|_{\frac{3\pi}{2}}^{2\pi}$$

$$A = (2) + \frac{1}{\pi} \left\{ (16\pi^2 - 30\pi^2) - \left(9\pi^2 - \frac{45\pi^2}{2} \right) \right\}$$

$$A = 2 + \frac{1}{\pi} \left\{ -14\pi^2 - 9\pi^2 + \frac{45\pi^2}{2} \right\}$$

$$A = 2 + \frac{1}{\pi} \left(-23\pi^2 + \frac{45\pi^2}{2} \right)$$

$$A = 2 + \frac{1}{\pi} \left(-\frac{\pi^2}{2} \right)$$

$$A = 2 - \frac{\pi}{2}$$

2. Ans (A,D)

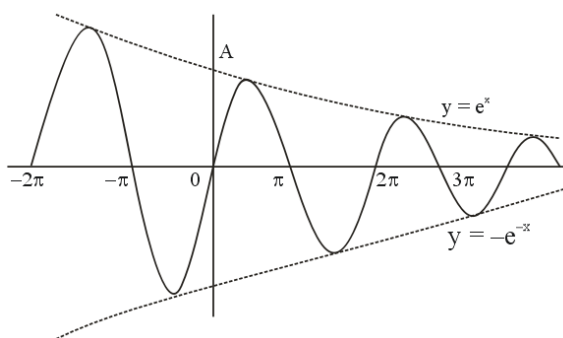
As we know $-1 \leq \sin x \leq 1$

$$y = e^{-x} \sin x$$

$$\Rightarrow -e^{-x} \leq y \leq e^{-x}$$

then y is bounded between

$$y = -e^{-x} \text{ and } y = e^{-x}$$



(A) Area bounded by curve and x-axis

between $x = -2\pi$ and $x = 0$

$$A = \int_{-2\pi}^{-\pi} e^{-x} \sin x \, dx + \left| \int_{-\pi}^0 e^{-x} \sin x \, dx \right|$$

$$A = \left(\frac{-e^{-x}(\sin x + \cos x)}{2} \right)_{-2\pi}^{-\pi} + \left| \left(\frac{-e^{-x}(\sin x + \cos x)}{2} \right)_{-\pi}^0 \right|$$

$$= \left(\frac{e^{\pi} + 1}{\sqrt{2}} \right)^2$$

(B) Area bounded by curve and x axis between

ordinate $x = -\pi$ and $x = \pi$

$$A = \left| \int_{-\pi}^0 f(x) \, dx \right| + \int_0^{\pi} f(x) \, dx$$

$$A = \frac{e^{\pi} + e^{-\pi} + 2}{2}$$

from graph we can say that option (C) is not

always true but option (D) is always true

3. Ans (A,C,D)

$$y^2 (x \cdot e^y \, dy - e^y \cdot dx - x \cdot dx)$$

$$+ xy \left(x \, dy - \left(\frac{dy}{y} + \frac{dx}{x} \right) \right) = x^2 y^2$$

$$\frac{x \cdot e^y \, dy - e^y \, dx}{x^2} + \frac{x \, dy - y \, dx}{xy} - \frac{(x \, dy + y \, dx)}{x^2 y^2} = 1$$

$$d \left(\frac{e^y}{x} \right) + d \left(\ln \left(\frac{y}{x} \right) \right) + d \left(\frac{1}{xy} \right) = 1$$

$$\frac{e^y}{x} + \ln \left(\frac{y}{x} \right) + \frac{1}{xy} = x + C$$

$$\therefore y(1) = 1$$

$$\frac{e}{1} + 0 + 1 = 1 + C$$

$$C = e$$

$$\frac{e^y}{x} + \ln \left(\frac{y}{x} \right) + \frac{1}{xy} = e + x$$

$$\frac{1}{x} \left(e^y + \frac{1}{y} \right) + \ln \left(\frac{y}{x} \right) = e + x$$

$$h(y) = e^y + \frac{1}{y}$$

$$g(x, y) = \frac{y}{x}$$

$$h(2) = \frac{2 \cdot e^2 + 1}{2}$$

$$h'(x) = e^x - \frac{1}{x^2} > 0, \forall x \in [1, \infty)$$

$$\Rightarrow \frac{y}{x} = 5$$

$\Rightarrow y = 5x$ represent a st. line passing through origin

4. Ans (A,C,D)

$$F(\lambda) = \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^{\lambda}}{\left(\sum_{r=1}^n r^4 \right) \left(\sum_{r=1}^n r^3 \right)}$$

$$= \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^{\lambda} \right) \cdot n^{\lambda+1}}{\left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^4 \right) \left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^3 \right) \cdot n^9}$$

$$= \lim_{n \rightarrow \infty} \frac{\int_0^1 x^{\lambda} dx}{\left(\int_0^1 x^4 dx \right) \left(\int_0^1 x^3 dx \right)} \cdot n^{\lambda-8}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{20}{(\lambda+1)} \cdot \frac{1}{n^{8-\lambda}} = F(\lambda)$$

5. Ans (C,D)

$$y = x^2 - 3 \quad \dots(1)$$

$$y = kx + 2 \quad \dots(2)$$

solving (1) & (2)

$$x^2 - kx - 5 = 0$$

Let α and β are roots of this then

$$\alpha + \beta = k \text{ and } \alpha\beta = -5$$

Required area

$$\begin{aligned} A &= \left[\int_{\alpha}^{\beta} (x^2 - 3) - (kx + 2) dx \right] \\ &= \left[\frac{x^3}{3} - \frac{kx^2}{2} - 5x \right]_{\alpha}^{\beta} \\ &= \left| \frac{1}{3} [\beta^3 - \alpha^3] - \frac{k}{2} [\beta^2 - \alpha^2] - 5(\beta - \alpha) \right| \\ A &= \left| (\beta - \alpha) \left(\frac{\alpha^2 + \beta^2 + \alpha\beta}{3} - \frac{k}{2}(\beta + \alpha) - 5 \right) \right| \end{aligned}$$

Putting the values of $\alpha + \beta$ and $\alpha\beta$

$$A = \frac{1}{6} (k^2 + 20)^{\frac{3}{2}}$$

for least value of A, $k = 0$

$$\text{Least value of } A = \frac{1}{6} (20)^{\frac{3}{2}} = \frac{20\sqrt{5}}{3}$$

6. Ans (A,B)

Since $x = 0$ is critical point of $f(x)$ but $f(x)$ does not have local extrema at $x = 0$

$$\text{so } f'(0) = 0 \text{ and } f''(0) = 0$$

$$\Rightarrow f'(x) = a(x - 0)^2$$

$$f(x) = \frac{ax^3}{3} + C$$

$$\Rightarrow f(0) = 1, f(1) = 2$$

$$f(x) = x^3 + 1$$

$$\Rightarrow \int \frac{x^3 + 1}{\sqrt{x^2 + 7}} dx = \int \frac{x^3 dx}{\sqrt{x^2 + 7}} + \int \frac{dx}{\sqrt{x^2 + 7}}$$

$$\Rightarrow \frac{(x^2 + 7)^{\frac{3}{2}}}{3} - 7(x^2 + 7)^{\frac{1}{2}} + \log(x + \sqrt{x^2 + 7})$$

$$\Rightarrow \frac{\sqrt{x^2 + 7}}{3} [x^2 - 14] + \log(x + \sqrt{x^2 + 7}) + C$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

$$\int_0^{f(x)} f^{-1}(t) dt = \int_0^x (\cos(t) - f(t)) dt$$

$$f^{-1}(f(x)) \cdot f'(x) = \cos x - f(x)$$

$$xf'(x) = \cos x - f(x)$$

$$xf'(x) + f(x) = \cos x$$

$$\frac{d}{dx} [xf(x)] = \cos x$$

$$xf(x) = \sin x + c$$

$$\text{but } f(0) = 1, c = 0$$

$$\Rightarrow f(x) = \frac{\sin x}{x}$$

$$\begin{vmatrix} f(2x) & f(x) \\ \sin x & 2 \end{vmatrix} = 0$$

$$2f(2x) - f(x) \cdot \sin x = 0$$

$$2 \cdot \frac{\sin 2x}{2x} - \frac{\sin x}{x} \cdot \sin x = 0$$

$$\sin x [2 \cos x - \sin x] = 0$$

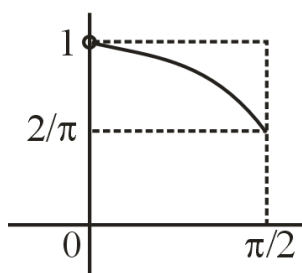
$$\text{For } x \in (0, 2\pi)$$

$$\sin x = 0, \Rightarrow x = \pi$$

$$\tan x = 2 \quad \text{two solution}$$

$$\text{total number of solutions} = 3$$

8. Ans (D)



$$1 < \int_0^{\pi/2} \frac{\sin x}{x} < \frac{\pi}{2}$$

9. Ans (C)

$$\text{Let } \int_0^{\pi} \frac{x}{1 - \cos \phi \cdot \sin x} dx = f(\pi - \phi) \quad \dots(1)$$

$$I = \int_0^{\pi} \frac{\pi - x}{1 - \cos \phi \cdot \sin(\pi - x)} dx \quad \dots(2)$$

Adding (1) & (2)

$$2I = \pi \cdot \int_0^{\pi} \frac{dx}{1 - \cos \phi \cdot \sin x}$$

$$I = \frac{\pi}{2} \cdot \int_0^{\pi} \frac{dx}{1 - \cos \phi \cdot \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}}$$

$$= \frac{\pi}{2} \int_0^{\pi} \frac{\sec^2 \frac{x}{2} \cdot dx}{1 + \tan^2 \frac{x}{2} - 2 \cos \phi \cdot \tan \frac{x}{2}}$$

Putting $\tan \frac{x}{2} = t$

$$\frac{1}{2} \sec^2 \frac{x}{2} \cdot dx = dt$$

$$I = \pi \int_0^{\infty} \frac{dt}{1 + t^2 - 2t \cos \phi}$$

$$= \pi \int_0^{\infty} \frac{dt}{(t - \cos \phi)^2 + \sin^2 \phi}$$

$$= \pi \left[\frac{1}{\sin \phi} \cdot \tan^{-1} \left(\frac{t - \cos \phi}{\sin \phi} \right) \right]_0^{\infty}$$

$$I = \pi \frac{1}{\sin \phi} \left[\frac{\pi}{2} - (\tan^{-1} (-\cot \phi)) \right]$$

$$= \frac{\pi}{\sin \phi} \cdot \left[\frac{\pi}{2} + \frac{\pi}{2} - \phi \right]$$

$$= \frac{\pi(\pi - \phi)}{\sin \phi} = f(\pi - \phi) \text{ (As given)}$$

$$\Rightarrow f(x) = \pi \frac{x}{\sin x}$$

$$f\left(\frac{\pi}{6}\right) = \frac{\pi \cdot \frac{\pi}{6}}{\frac{1}{2}} = \frac{\pi^2}{3}$$

10. Ans (A)

$$\text{To find } \int_0^{\frac{\pi}{2}} \pi^2 \frac{x^2}{\sin^2 x}$$

$$= \pi^2 \int_0^{\frac{\pi}{2}} x^2 \cdot \operatorname{cosec}^2 x$$

$$= \pi^2 \left[(x^2(-\cot x))_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} 2x \cdot \cot x \, dx \right]$$

$$= \pi^2 \left[0 + (2x \cdot \ln \sin x)_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 2 \cdot \ln \sin x \, dx \right]$$

$$= \pi^2 \left[0 - 2 \left(-\frac{\pi}{2} \ln 2 \right) \right] = \pi^3 \ln 2$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (1.50)

$$f(x) = x^2 - \int_0^1 (f(t) + x)^2 dt$$

$$f(x) = x^2 - \int_0^1 (f^2(t) + x^2 + 2xf(t)) dt$$

$$= x^2 - \int_0^1 f^2(t) dt - x^2 - 2x \int_0^1 f(t) dt$$

$$f(x) = - \int_0^1 f^2(t) dt - 2x \int_0^1 f(t) dt$$

$$\text{Let } a = \int_0^1 f^2(t) dt \text{ and } b = \int_0^1 f(t) dt$$

$$\Rightarrow f(x) = -a - 2bx$$

$$a = \int_0^1 (-a - 2bt)^2 dt$$

$$a = a^2 + \frac{4b^2}{3} + \frac{4ab}{2} \quad \dots(1)$$

$$b = \int_0^1 (-a - 2bt) dt$$

$$b = -a - b$$

$$2b = -a \quad \dots(2)$$

$$a = a^2 + \frac{1}{3}a^2 + 2a\left(-\frac{a}{2}\right)$$

$$\Rightarrow a = 0 \text{ or } a = 3$$

$$a = 0 \text{ is not possible so } a = 3, b = -\frac{3}{2}$$

$$\Rightarrow f(x) = -3 + 3x = 3(x - 1)$$

$$a = \int_0^{\infty} \frac{\tan^{-1}x}{x(1+x^2)} dx$$

$$x = \tan\theta \quad a = \int_0^{\pi/2} \frac{\theta}{\tan\theta} d\theta$$

$$= [\theta \ln \sin \theta]_0^{\pi/2} - \int_0^{\pi/2} \ln \sin \theta d\theta$$

$$a = \frac{\pi}{2} \ln 2$$

$$b = \int_0^1 \frac{5x^8 + 3x^4 + 1}{\sqrt{x^8 + x^4 + 1}} dx$$

$$= \int_0^1 \frac{5x^9 + 3x^5 + x}{\sqrt{x^{10} + x^6 + x^2}} dx$$

$$x^{10} + x^6 + x^2 = t$$

$$(10x^9 + 6x^5 + 2x) dx = dt$$

$$(5x^9 + 3x^5 + x) dx = \frac{dt}{2}$$

$$b = \int_0^3 \frac{dt}{2\sqrt{t}} = \frac{1}{2} \left[\frac{t^{1/2}}{1/2} \right]_0^3 = \sqrt{3}$$

For Question 11

$$\int_1^3 3(x-1)dx = 3 \left(\frac{x^2}{2} - x \right) \Big|_1^3$$

$$= 3 \left(\frac{9}{2} - 3 - \frac{1}{2} + 1 \right) = 3(4 - 2) = 6$$

2. Ans (4.50)

$$c = \int_0^2 \sqrt{x^3 + 1} dx + \int_1^3 (x^2 - 1)^{\frac{1}{3}} dx$$

$$f(x) = (x^3 + 1)^{\frac{1}{2}}, f^{-1}(x) = (x^2 - 1)^{\frac{1}{3}}$$

$$c = 2f(2) - 0f(0) = 2(9)^{\frac{1}{2}} = 6$$

$$d = \int_0^1 \frac{3x^2 + 9x + 7}{(x^2 + 3x + 2)^3} dx$$

$$= \int_0^1 \frac{3x^2 + 9x + 7}{(x+2)^3(x+1)^3} dx$$

$$= \int_0^1 \frac{1}{(x+1)^3} - \frac{1}{(x+2)^3} dx$$

$$= \frac{1}{2} \left(\frac{1}{(x+2)^2} - \frac{1}{(x+1)^2} \right) \Big|_0^1$$

$$d = \frac{11}{36}$$

For Question 12

$$y = 3x - 3$$

$$x = \frac{y+3}{3} \Rightarrow f^{-1}(x) = \frac{x+3}{3} = \frac{x}{3} + 1$$

$$A = \int_2^6 \left(\frac{x}{3} + 1 \right) dx = \left(\frac{x^2}{6} + x \right) \Big|_2^6$$

$$= \frac{28}{3}$$

3. Ans (2.00)

$$\int \frac{dx}{\sqrt{a^{5x}} \sqrt[4]{(a^{2x} + a^{-2x})^3}}$$

(where $t = \sqrt{2 + \sin^{1/3} x}$)

$$= \frac{1}{\ln a} \int \frac{dt}{t \cdot t^{5/2} \{t^2 + t^{-2}\}^{3/4}}$$

$$\Rightarrow a^x = t \Rightarrow a^x \ln a \cdot dx = dt$$

$$= \frac{1}{\ln a} \int \frac{dt}{t^{7/2} \frac{(1+t^4)^{3/4}}{t^{3/2}}}$$

$$\left(\because \frac{m+1}{n} - \frac{p}{q} - \text{integer} \frac{-2+1}{4} - \frac{3}{4} = -1 \right)$$

$$= \frac{1}{\ln a} \int \frac{dt}{t^2 (1+t^4)^{3/4}}$$

$$\frac{1}{t^4} = 1 = u^4 \Rightarrow \frac{1}{t^5} dt = 4u^3 du$$

$$= \frac{1}{\ln a} \int \frac{-u^3 du}{u^3}$$

$$= -\frac{1}{\ln a} \left\{ \frac{1}{t^4} + 1 \right\}^{\frac{1}{4}} + C$$

$$= \frac{1}{\ln a} \{a^{-4x} + 1\}^{\frac{1}{4}} + C$$

4. Ans (8.00)

$$I = \int \sqrt{\operatorname{cosec} x} \left(2 \sqrt[3]{\operatorname{cosec} x} + 1 \right)^{\frac{3}{2}} \cdot \cos x dx$$

$$= \int \frac{(2 + \sqrt[3]{\sin x})^{\frac{3}{2}}}{\sin^{1/3} x} \cdot \frac{\cos x}{\sin^{2/3} x} dx$$

Let

$$2 + \sin^{1/3} x = t^2$$

$$\frac{1}{3} \frac{\cos x}{\sin^{2/3} x} dx = 2t dt$$

$$I = \int \frac{t^3 \cdot 6t}{t^2 - 2} dt = 6 \int \frac{t^4 - 4 + 4}{t^2 - 2} dt$$

$$= 6 \int \left[t^2 + 2 + \frac{4}{t^2 - 2} \right] dt$$

$$= 6 \left[\frac{t^3}{3} + 2t + \frac{4}{2\sqrt{2}} \ln \left| \frac{t - \sqrt{2}}{t + \sqrt{2}} \right| \right] + C$$

$$= 2t^3 + 12t + 6\sqrt{2} \ln \left| \frac{t - \sqrt{2}}{t + \sqrt{2}} \right| + C$$

$$\text{where } t = \sqrt{2 + \sin^{1/3} x}$$

5. Ans (261.00)

$$\text{Let } f(x) = ax^3 + bx^2 + cx + d$$

It is passing through origin then $d = 0$

$$f'(x) = 3ax^2 + 2bx + c$$

$$f''(x) = 6ax + 2b$$

$$f''(0) = 0 + 2b = 0 \Rightarrow b = 0$$

$$f' \left(\frac{-1}{\sqrt{10}} \right) = 0$$

$$\Rightarrow 3a \left(\frac{-1}{\sqrt{10}} \right)^2 + 0 + c = 0$$

$$3a + 10c = 0 \quad \dots(i)$$

$$f(1) = 7 \Rightarrow a + c = 7 \quad \dots(ii)$$

from (i) and (ii)

$$a = 10 \text{ and } c = -3$$

$$f(x) = 10x^3 - 3x$$

$$f(3) = 10 \cdot 3^3 - 3 \times 3 = 270 - 9 = 261$$

6. Ans (14.00)

$$\frac{dy}{dx} + f'(x) \cdot y = (f(x) + 1) f'(x)$$

$$\text{IF} = e^{\int f'(x) dx} = e^{f(x)}$$

$$y \cdot e^{f(x)} = \int e^{f(x)} (f(x) + 1) f'(x) dx$$

$$\text{Let } f(x) = t$$

$$f'(x) dx = dt$$

$$y \cdot e^t = \int e^t (t + 1) dt$$

$$y \cdot e^t = t \cdot e^t + C$$

$$g(x) e^{f(x)} = f(x) e^{f(x)} + C$$

$$g(x) e^{10x^3 - 3x} = (10x^3 - 3x) e^{(10x^3 - 3x)} + C$$

Put $x = 0$

$$1 \cdot e^0 = 0 + c \Rightarrow c = 1$$

$$g(1) e^7 = 7 \cdot e^7 + 1$$

$$g(1) = 7 + e^{-7} = \alpha + e^{-\beta}$$

$$\Rightarrow \alpha + \beta = 14$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (12)

$$\text{Circle } x^2 + y^2 - 4x + 2y + 1 = 0$$

$$\text{or } (x-2)^2 + (y+1)^2 = 4$$

$$\text{So } x \in [0, 4]$$

$$0 \leq \frac{x^3}{101} + \frac{x}{36} < 1, \quad \forall x \in [0, 4]$$

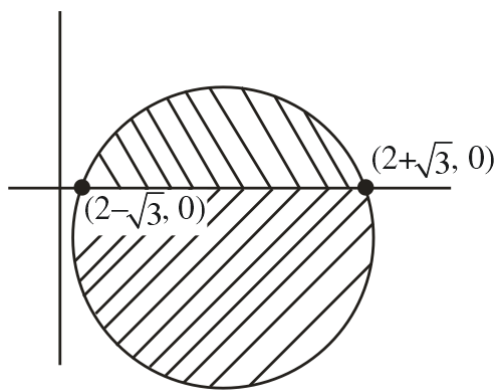
$$\left[\frac{x^3}{101} + \frac{x}{36} \right] = 0 \Rightarrow y = 0$$

Now at x-axis $y = 0$

$$(x-2)^2 = 3$$

it cuts the x-axis at points $(2 - \sqrt{3}, 0)$ and

$$(2 + \sqrt{3}, 0)$$



Let

$$A = \int_{2-\sqrt{3}}^{2+\sqrt{3}} \left(\sqrt{4 - (x-2)^2} - 1 \right) dx$$

$$= \frac{4\pi - 3\sqrt{3}}{3}$$

The required ratio

$$= \frac{A}{4\pi - A} = \frac{4\pi - 3\sqrt{3}}{8\pi + 3\sqrt{3}}$$

8. Ans (2)

$$x \frac{dy}{dx} + y = \frac{2 \cos x + 3}{(2 + 3 \cos x)^2}$$

$$\frac{dy}{dx} + \frac{1}{x} y = \frac{2 \cos x + 3}{x(2 + 3 \cos x)^2}$$

$$\text{IF} = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

$$y \cdot x = \int x \cdot \frac{\cos x + 3}{x(2 + 3 \cos x)^2} dx$$

$$y \cdot x = \frac{\sin x}{2 + 3 \cos x} + C$$

$$f\left(\frac{\pi}{2}\right) = \frac{2}{\pi}$$

$$\frac{2}{\pi} \cdot \frac{\pi}{2} = \frac{1}{2} + C \Rightarrow C = \frac{1}{2}$$

$$f(x) = \frac{1}{x} \left[\frac{\sin x}{2 + 3 \cos x} \right] + \frac{1}{2x}$$

$$\pi \cdot f\left(\frac{\pi}{3}\right) = \left[\frac{\frac{\sqrt{3}}{2}}{2\left(2 + \frac{3}{2}\right)} + \frac{1}{2} \right] 3$$

$$\pi \cdot f\left(\frac{\pi}{3}\right) = \left(\frac{2\sqrt{3} + 7}{14} \right) 3 = 2.24$$

$$[2.24] = 2 \text{ Ans.}$$

9. Ans (1)

$$\text{Let } g(x) = \int_0^x f(t) dt \Rightarrow g'(x) = f(x)$$

$$\lim_{x \rightarrow \infty} g(x) = \lim_{x \rightarrow \infty} \frac{e^x g(x)}{e^x}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x (g + g')}{e^x} = \lim_{x \rightarrow \infty} (g + g') = L$$

$$\therefore \lim_{x \rightarrow \infty} g(x) = \lim_{x \rightarrow \infty} (g + g' - g')$$

$$L = \lim_{x \rightarrow \infty} (g + g') - \lim_{x \rightarrow \infty} g'$$

$$\Rightarrow \lim_{x \rightarrow \infty} g'(x) = 0$$

$$\Rightarrow \lim_{x \rightarrow \infty} f(x) = 0$$